CS19541 – COMPUTER NETWORKS LAB

PRACTICAL – 10

INTERNETWORKING USING ROUTERS AND A WIRELESS ROUTER IN CISCO PACKET TRACER

Aim

To set up internetworking with routers and a wireless router in Cisco Packet Tracer and check communication between the connected devices.

Requirements

Cisco Packet Tracer, routers, desktop PCs, laptop, wireless router, cable modem, Internet cloud and server.

PART A – ROUTER-BASED INTERNETWORKING

Configure Router1 to connect two separate networks. Assign FastEthernet0/0 the address 192.168.10.1/24 and FastEthernet0/1 the address 192.168.20.1/24. Set PC0 to 192.168.10.2 with gateway 192.168.10.1, and PC1 to 192.168.20.2 with gateway 192.168.20.1.

Router Configuration Commands

```
enable
configure terminal
interface fastethernet0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
interface fastethernet0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
end
```

PART B – WIRELESS ROUTER AND DHCP

Set the wireless network name (SSID) as HomeNetwork and turn on DHCP. Enter 208.67.220.220 as the DNS server. Connect the laptop to the wireless network and configure the PC to receive its IP address automatically. On the server, set up the DHCP pool with starting IP address 208.67.220.1, subnet mask 255.255.255.0, DNS address 208.67.220.220 and a maximum of 50 users.

Procedure

1. Add all required network devices to the Packet Tracer workspace.
2. Connect the PCs to the router using copper straight-through cables.
3. Configure the router interfaces and the PCs' IP addresses and gateways.
4. Create the wireless topology and configure the SSID and DHCP settings.
5. Connect the laptop to the wireless network.
6. Test communication using ping or the Simple PDU tool.

Input

IP addresses, subnet masks, default gateways, wireless SSID and DHCP configuration values.

Output

Connected devices obtain the assigned network settings and can communicate across the configured networks. Use ping or Simple PDU to verify connectivity.

Observation

The router interfaces were assigned IP addresses and enabled. The PCs were configured with their respective IP addresses and default gateways. The wireless network and DHCP options were set, and communication was tested.

Result

Thus, internetworking with routers and a wireless router was configured and tested successfully in Cisco Packet Tracer.